

2.1.4 Enzymes

Metabolism in living organisms relies upon enzyme-controlled reactions. Knowledge of how enzymes function and the factors that affect enzyme action has

improved our understanding of biological processes and increased our use of enzymes in industry.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the role of enzymes in catalysing reactions that affect metabolism at a cellular and whole organism level	To include the idea that enzymes affect both structure and function.
(b) the role of enzymes in catalysing both intracellular and extracellular reactions	To include catalase as an example of an enzyme that catalyses intracellular reactions and amylase as an example of an enzyme that catalyses extracellular reactions.
(c) the mechanism of enzyme action	To include the tertiary structure, specificity, active site, lock and key hypothesis, induced-fit hypothesis, enzyme-substrate complex, enzyme-product complex, product formation and lowering of activation energy. HSW1, HSW8
(d) (i) the effects of pH, temperature, enzyme concentration and substrate concentration on enzyme activity	To include reference to the temperature coefficient (Q_{10}).
(ii) practical investigations into the effects of pH, temperature, enzyme concentration and substrate concentration on enzyme activity	$Q_{10} = \frac{R_2}{R_1}$ An opportunity for serial dilutions. <i>M0.1, M0.2, M0.3, M1.1, M1.3, M1.11, M3.1, M3.2, M3.3, M3.5, M3.6</i> PAG4 HSW1, HSW2, HSW4, HSW5, HSW6, HSW8.
(e) the need for coenzymes and cofactors in some enzyme-controlled reactions	To include the chloride ion as a cofactor for amylase and vitamins as a source of coenzymes. PAG4
(f) the effects of inhibitors on the rate of enzyme-controlled reactions.	To include competitive and non-competitive, reversible and non-reversible inhibitors, and the role of end-product inhibition. <i>M0.1, M0.2, M0.3, M1.1, M1.3, M1.11, M3.1, M3.2, M3.3, M3.5, M3.6</i> PAG4 HSW1, HSW2, HSW4, HSW5, HSW6, HSW8